

Sensorimotor-Aligned Design for Pareto-Efficient Haptic Immersion in XR

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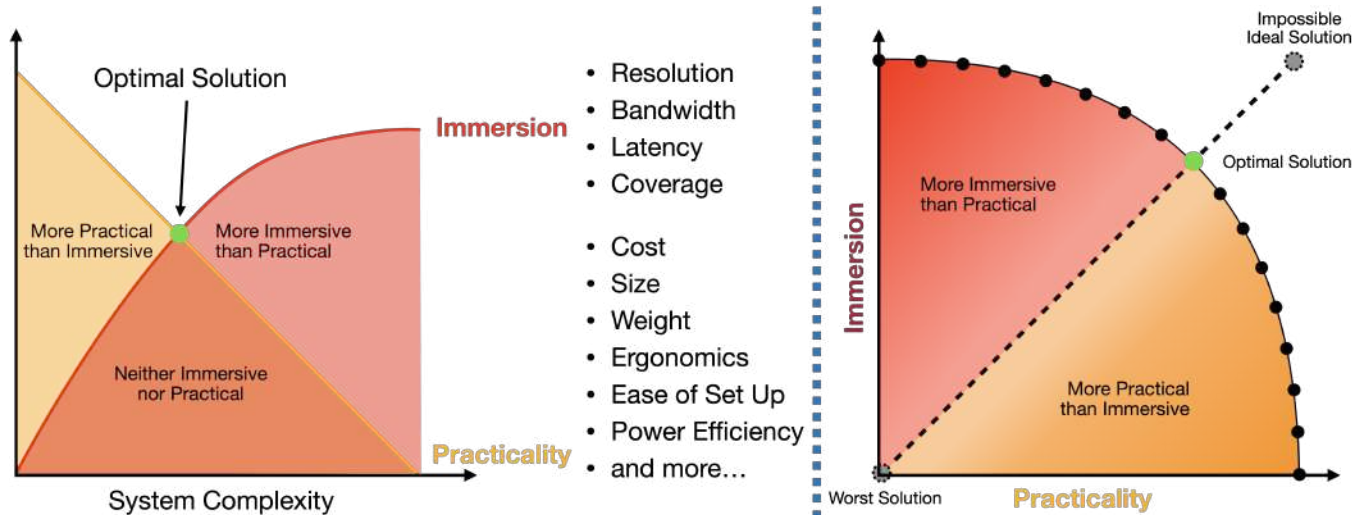


Figure 1: Haptics design framework optimizing across two objectives. Left: The design ideals of immersion and practicality compared to system complexity, which results in an inverse relationship. Right: Immersion and practicality represented as opposing axes, with the optimal Pareto-efficient solution of my approach outlined, maximally optimizing both objectives.

Abstract

Augmented and virtual reality headsets, collectively referred to as extended reality (XR), can alter, augment, or even replace our reality. While these headsets have made impressive strides in audio-visual immersion over the past half-century, XR interactions remain almost completely absent of appropriately expressive tactile sensations. At present, even mainstream consumer systems rely on vibrotactile haptic actuators in the controllers, which are inherently limited to clicks and buzzes — an exceedingly limited range of expressivity with which to represent the rich tactile world.

Realizing the holistic promise of XR requires full-body haptic immersion, just as much as it requires full audio-visual immersion. To frame critical design considerations in haptics research, I propose an immersion-practicality tradeoff model. These competing objectives underscore the inherent tension between providing rich

sensory feedback (often e.g. costly, bulky), while maintaining consumer feasibility and usability (e.g., low cost, easy to use). Under this framework, I sought to identify and build Pareto-efficient haptic systems where the haptic approach is aligned with humans' sensorimotor system. I present three published projects that embody my design approach with tactile haptics to different regions of the body. My proposed future work extends this framework to the other major dimensions of haptics (kinesthetic/force feedback). These systems serve as probes into my research approach and advance the vision of practical, immersive, full-body haptics.

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1 Introduction

The boom in commercial virtual and augmented reality has led to headsets that expand our reality or even fabricate new ones, enabling capabilities like realistic remote collaboration or immersive gaming experiences. The core technologies on these headsets are

impressive — high resolution, low latency visual displays and spatial audio systems. These technologies enable amazing audio-visual capabilities, enhancing our senses in XR.

However, the sense of touch has remained largely underdeveloped. Commercial devices rely on vibrotactile elements embedded in controllers, but vibrations can only represent a very limited sample of tactile sensations within the diverse haptic dimensions. For example, common everyday haptic sensations include forces (like weight), surface textures, compliance, thermal, and more, none of which can be accurately represented with vibration alone. Additionally, haptic systems must adapt to this paradigm of full immersion, requiring full-body tactile sensations to match powerful audio-visual surroundings. When a user feels the ocean breeze on a virtual beach, they wouldn't expect to feel it only on their hands, but also replicated accordingly across the entire body. This contextualizes the complexity of building haptic devices for XR — we want full-body, realistic tactile immersion.

2 Proposed Research Approach: Sensorimotor-Aligned Design for Pareto-Efficient Haptic Immersion in XR

To inform design choices for new haptic systems research, I propose a new multi-object optimization problem (MOOP) framework with two critical design objectives: **immersion** and **practicality**.

2.1 Pareto-Efficient Haptic Immersion

When designing a haptic system, we want it to be **practical** enough to be used but also **immersive** enough to be worth using. The idea of immersion can be broken down generally into parameters like resolution (density of output over space); bandwidth (speed of rendering); latency (delay in the system); and coverage (how much of the body covered). By graphing the relationship of immersion to system complexity (Figure 1, left) we see that typically, as a system gets more complex, it is able to offer more immersion as more resources are added. This isn't directly a linear relationship, because at some point our sense of touch is oversaturated and added complexity will not lead to added benefits for immersion. Practicality is simpler to understand, and includes factors such as cost, size, weight, ergonomics, ease of setup, power efficiency, and more. The more practical a system is, the easier it is for a user to just pick up and use it. Practicality has an inverse relationship to immersion (Figure 1, left); as the complexity of a system goes up, it typically gets more impractical. When graphed together, we can see that the left side of the intersection are systems that are more practical but less immersive, whereas systems on the right side are more immersive but less practical.

This inverse relationship can be represented by rearranging the graph to make practicality and immersion opposing axes (Figure 1, right). Here we outline the Pareto front, which are all solutions that have an efficient trade-off between the two objectives, known as Pareto-efficient solutions. For example, at the very top are systems that are fully immersive but completely impractical, like an expensive, heavy full body exosuit with tons of actuators. At the bottom are systems that are incredibly practical, but with no haptic immersion at all, like bare hand input. By weighing the two design objectives equally, the hypothetical ideal solution is completely

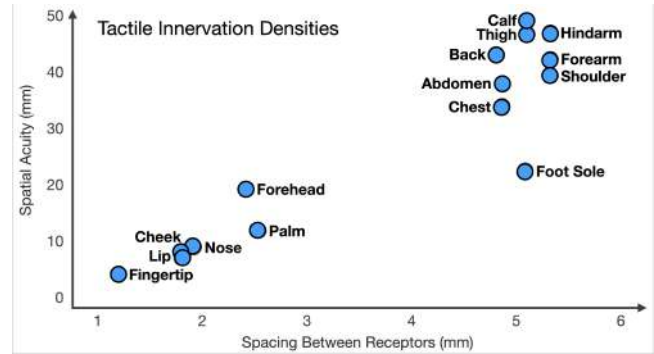


Figure 2: Tactile innervation densities across the whole body [1]. The closer to the bottom left corner, the more sensitive.

immersive and maximally practical, but in reality this would not be possible. Drawing a line to this impossible ideal solution intersects our optimal solution, which is the closest Pareto-efficient solution, where both are maximally optimized and no improvements can be made to one objective without making the other worse.

2.2 Sensorimotor-Alignment

Optimal Pareto-efficiency over the whole body is further complicated because our skin does not perceive tactile stimulation equally. In 2020, Corniani et al. [1] studied tactile innervation densities (Figure 2). The lower the spacing between receptors means the higher the density, and a high density corresponds to more sensitivity. Essentially, the bottom left corner of the graph outlines the most sensitive regions of our body. Additionally, Wilder Penfield mapped the areas of the brain that correlated with both our motor and sensory cortices [5, 6] through electrical stimulation. This led to the development of the somatosensory homunculus (Figure 3) — in this visual representation, the size of the body part represents its tactile sensitivity, aligning closely with Corniani.

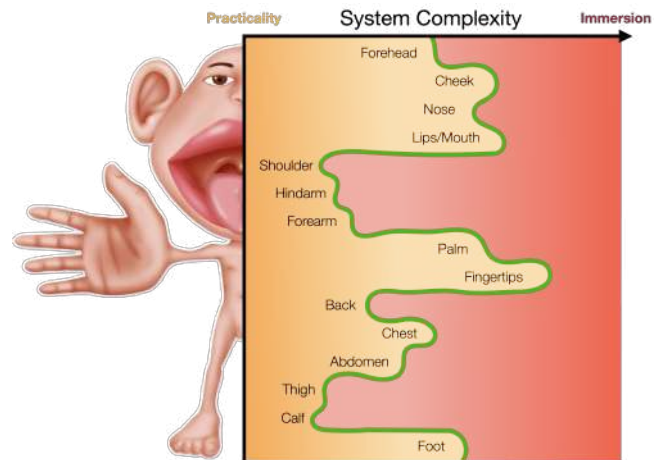


Figure 3: The sensory homunculus visually represents body part sensitivity through size, with the corresponding practicality-immersion tradeoffs to the right.

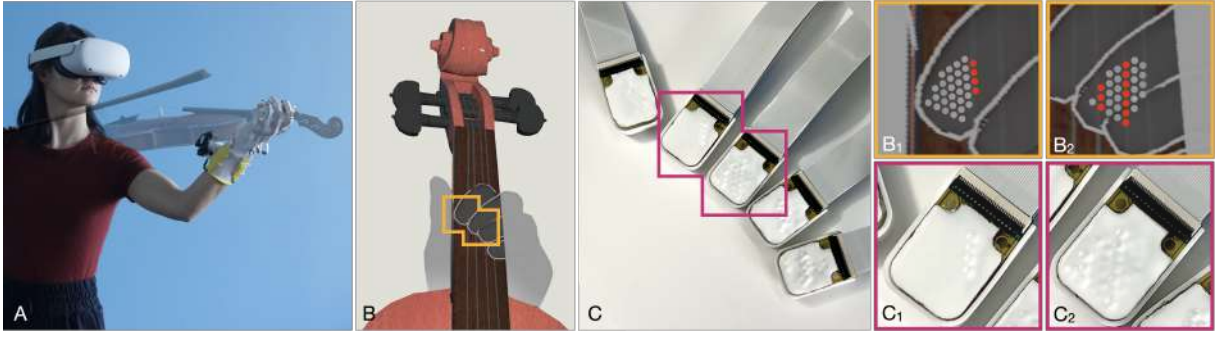


Figure 4: Our self-contained, wireless, Fluid Reality gloves enable high-resolution haptic immersion in AR/VR applications (A & B). Here, a user feels the individual strings of a virtual violin, rendered appropriately across five fingertip haptic arrays (C).

In Figure 3, on the right of the homunculus, we outline a continuous curve to represent relative tactile sensitivities. Our ideal full-body system should fit this curve like a jigsaw piece, optimizing for maximal immersion-practicality tradeoff per body region. This is my research approach: Sensorimotor-Aligned Design for Pareto-Efficient Haptic Immersion in XR.

3 High tactile sensitivity of the hands [Fluid Reality, UIST'23]

Our main tactile interactions utilize our hands, the most sensitive parts of our bodies. Commercial XR haptic solutions often focus on vibrating the hands with controllers, which cannot convey the rich depth of tactile information that our hands can sense. A new approach was necessary to sufficiently stimulate our fingertips without compromising the ease of use.

In this paper [9], we developed a significant advancement in the state of the art for haptic gloves (Figure 4). The tactile output layer of our system is broadly similar to commercial prototypes (i.e. from HaptX, Meta) – a deformable 2D arrangement of haptic cell "pixels". However, the underlying technical approach is radically different. We leverage small embedded electroosmotic pumps (EOPs) (as explored by Shultz and Harrison [11]) to enable microfluidic deformations on the fingertips. Although gloves from HaptX and Meta are similar in capability to ours, their gloves must

be tethered to bulky external valves and pumps, making them infeasible for dynamic use. Our approach (wireless or USB tethered) can generate high pressures and fast dynamic flows, offering dramatic miniaturization (several orders of magnitude) while also reducing power consumption, weight, and cost. We developed a variety of expressive building blocks for rich virtual scenes, including real-time haptic rendering of surface textures, compliance, and general tactile shapes.

This new approach for creating high-density wearable haptic arrays unlocked the possibility of completely untethered yet high-resolution stimuli at the fingertips. Importantly, the lack of external infrastructure drastically increases the practicality of the system, while retaining the capability for high bandwidth, expressive, and real-time haptics.

4 High tactile sensitivity of the lips [Mouth Haptics, CHI'22 BEST PAPER]

The mouth is largely overlooked as a haptic target, despite being second in density of mechanoreceptors (Figure 2). However, consumers do not want to cover their entire face, let alone put something against (or into) their mouth. Instead, the proximity of the headset to the mouth offers a significant opportunity to enable on- and in-mouth haptic effects.



Figure 5: Our system delivers rich, non-contact haptic feedback to the mouth using a thin array of ultrasonic transducers integrated into the underside of a VR headset. A user leans forward to drink from a virtual water fountain, feeling the stream of water on their lips and teeth.

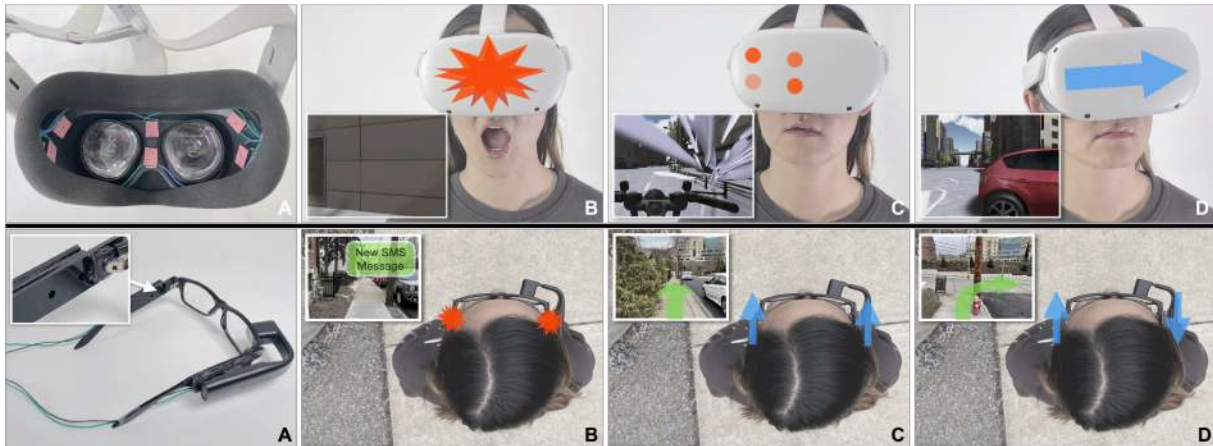


Figure 6: Top: VR headset with six small synjets (A). Effects include a large simultaneous impulse for impact (B), random pulses representing water drops (C), and sequential firing of the array for a swipe effect, like wind (D). Bottom: AR headset with two dual-port synjets (A), enabling notification impulses (B) and wayfinding swipes (C & D)

In this research [10], we built a thin, compact, beam forming array of ultrasonic transducers, fitted to the bottom of the headset without contacting or obstructing the user’s mouth (Figure 5). This device focuses air-borne acoustic energy onto the lips to create sensations such as taps and vibrations, which can also be animated along arbitrary 3D paths. In addition to the lips, our effects can be felt on the teeth and tongue. When coupled with coordinated graphical feedback, we evaluated that the effects were convincing, boosting realism and immersion. We built a variety of sensory demos, including raindrops, mud splatter, pushing through cobwebs, and crawling bugs. We were the first to integrate haptic ultrasonic phased arrays into a headset for use on the mouth and explore the rich application space.

Our Mouth Haptics device enables acutely spatially localized expressivity not previously possible in prior work (e.g. diffuse wind from fans [2, 7]). Our method also reduces weight, power consumption, and latency in comparison to [13]. Importantly, our device leaves the mouth exposed and integrates directly onto the bottom of the headset, allowing for easy usability and a hygienic form factor. In Mouth Haptics, we demonstrate these practicality improvements while preserving somatosensory-aligned high-fidelity immersion, adhering to the Pareto-efficient design framework.

5 Medium tactile sensitivity of the face [Synjets, TOCHI’23 & CHI’24]

Right behind the mouth and the fingertips in sensitivity is the rest of the face - namely, the cheek, nose, and forehead. These regions are often semi or fully covered by modern-day VR headsets, and directly contacting the facial cup. Prior work experimented with embedding various outputs directly into this lining [3, 4, 12]. Our approach was to instead instrument the inside of the facial cup, allowing access to more surface area of the face.

We accomplish this with non-contact, "mid-air" haptic technology. Only a handful of mid-air actuation methods existed in the literature, with various strengths and weaknesses. In this paper [8], we explored an emerging haptic approach built around synthetic

jets (or "synjets") – simple devices capable of generating a zero-net-mass-flux jet of air. Synjets are highly controllable by adjusting parameters such as windspeed, peak pressure, and jet diameter to create temporal and spatial effects. The range and impact forces of synjets are comparatively large, offering more salient haptics than similar techniques such as fans or ultrasound phased arrays. Finally, because synjets are highly scalable, we were able to build centimeter-scale example devices; the VR headset (Figure 6, top) had six small synjets within the facial cavity, while the AR glasses had a dual-ported synjet on each side at the temple (Figure 6, bottom).

While synjets compromise on expressivity when compared to direct contact methods, the simplicity and practicality of the method makes them suitable for our less sensitive facial skin. In our example devices, the synjets cost only \$2, with a diminutive power consumption of 172mW, allowing for easy integration into battery-powered devices like the VR headsets. Participants in our stimuli recognition user study were able to identify all 6 VR headset stimuli and all 5 AR glasses stimuli with an average accuracy of 97.7%, meaning that we can still render a broad set of distinguishable haptic stimuli using our synjets. Thus, this Pareto-efficient approach offers a significant increase in practicality while maintaining a similar level of expressive immersion to other mid-air haptic methods.

6 Low tactile, high kinesthetic sensitivity of the limbs [Proposed work; ongoing]

As seen in Figure 2, the rest of the body has much less spatial acuity. A high-resolution haptic system would not be Pareto-efficient for areas like our limbs and torso, as the extra fidelity would likely be a waste of resources at the cost of practicality. However, full body coverage is an important aspect of immersion, and we want to optimize our haptic actuation based on the perceptual capabilities of the limbs and torso.

To create this holistic experience and effectively deliver haptic complexity, we consider that our limbs are highly receptive to force output, due to their motor capabilities. As an investigation, we have constructed BLDC motor-driven reels of high-tensile string.

These are anchored to the user and pull on various parts of the body, providing varying force feedback. We propose creating a full-body suit with this easy, practical method, extending this method to connect our legs, arms, torso, and neck. The design of these reel units allows us to provide appropriate force resolution to many different body parts, creating sensorimotor-aligned immersion. We purposely choose not to use a direct drive system (with motors placed directly on the joints) that would have more torque and less latency, because the pliable string that transfers the motor torque can be designed into a soft exoskeleton. This makes the system more practical and easy to wear, as well as reduces fatigue from the weight on the user's extremities. We propose a suit with key controllable tug points that span the body, which reduces coverage but also reduces the number of motors needed and therefore the overall cost and size. This research will build toward this Pareto-efficient solution for full-body kinesthetic haptics through this reel-based soft exoskeleton.

7 Future Work and Open Questions

Most of my prior papers present a single, Pareto-efficient haptic system for a specific area of the body. The actuation methods and devices are uncorrelated, each with different immersion-practicality trade-offs, and could not be easily combined into one coherent system. Instead, they serve as probes into the sensorimotor-alignment design framework, and explore Pareto optimization with smaller goals. However, the ultimate goal is to have full-body tactile and kinesthetic haptics in extended reality that is both immersive and practical. While this may be achieved with various disparate devices, the most practical and unencumbered system would utilize a single method to realize this goal. This would improve practicality in ease of setup, but to not compromise on quality, the method would have to be scalable to varying perceptual sensitivities. Synjets [8] was an initial effort on this goal; scalable in range, area, and pressure to create variable mid-air haptics, but lacking in fidelity and expressivity. Additionally, my ongoing kinesthetic research seeks to enable scalable force feedback over the whole body, but cannot render any tactile haptics. An open question remains for how we can have fully Pareto-optimized haptic immersion over the whole body in XR, and I am excited to ideate and think deeper about this as I move forward in my research.

8 Dissertation Status

I am a fifth year Robotics PhD at Carnegie Mellon University, and I passed my thesis proposal on Nov. 4th. My projected dissertation completion is in late Fall 2025/early Spring 2026. I conduct research in the Future Interfaces Group of the Human Computer Interaction Institute, advised by Professor Chris Harrison. I have published two conference papers and one journal paper on my dissertation topic, and am currently working on two more related conference papers to be integrated into my final thesis defense. I've also published three non-thesis papers that focus more broadly on augmented interactions, with three more in submission. I've interned with a variety of industry research labs, including Disney Research, Meta Reality Labs, and Microsoft Research. My long-term goal is to seek a Research Scientist position in industry, broadening my scope to both output (haptics) and input (sensing) to create closed-loop

systems. I would love to continue to make an impact in improving the interaction experience in XR, enabling this new paradigm of technology to become both feasible and desirable.

9 Conclusion

This research statement presents my ongoing research in sensorimotor-aligned design for Pareto-efficient haptic immersion in XR. My approach to building haptic systems leverages the sensorimotor framework, representing the perceptual sensitivity of different body regions, to design haptic systems that are Pareto optimized for the immersion-practicality tradeoff. In my projects, I have developed practical, immersive systems for a variety of body locations, with haptics that are sufficient enough for the corresponding tactile receptivity to be immersive while still remaining feasible to use. I hope to dive deeper into implementations with modular haptic expressivity and adjustable resolution, honing my framework to serve as a generalizable design guide for haptic systems of the future.

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